

# Property (A) and Property (C) Do Not Imply Semi-Greediness

A literature identification for arXiv:1803.05052

Run `fa_banach_001`, lane 13

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## Source question

Berná, Dilworth, Kutzarova, Oikhberg, and Wallis ask in Section 6, Question 3 of *The weighted Property (A) and the greedy algorithm* (arXiv:1803.05052, PDF p. 19):

If a basis has Properties (C) and (A), is it necessarily semi-greedy?

Here the source works with a semi-normalized Schauder basis. Property (C) is the uniform estimate

$$\min_{j \in \Lambda} |e_j^*(x)| \left\| \sum_{j \in \Lambda} \varepsilon_j e_j \right\| \leq C \|x\|$$

for every greedy set  $\Lambda$  of  $x$  and every choice of signs. Property (A) is the usual symmetry-for-largest-coefficients condition.

**Answer.** *No. There exists a Schauder basis with Properties (A) and (C) that is not semi-greedy.*

## Decisive later result

P. M. Berná, *Equivalence between almost-greedy and semi-greedy bases* (arXiv:1804.05730), proves in Theorem 1.10 that a semi-greedy Schauder basis is quasi-greedy (and super-democratic). More precisely, if  $C_s$  is its semi-greedy constant and  $K_b$  its basis constant, the theorem gives a finite quasi-greedy constant, explicitly bounded there by

$$K_b(2 + 3(K_b C_s)^2).$$

On PDF p. 7, immediately before Section 4, the supporting paper explicitly returns to Question 3 of arXiv:1803.05052. It cites the example in Berná–Blasco–Garrigós, Subsection 5.5, which has Property (A) and the inequality defining Property (C), but is not quasi-greedy. By Theorem 1.10, that basis cannot be semi-greedy. Thus the cited example is a counterexample to the source question.

The logical implication is simply

$$\text{semi-greedy Schauder} \implies \text{quasi-greedy}, \quad \mathcal{B}_{\text{ex}} \text{ is not quasi-greedy},$$

while  $\mathcal{B}_{\text{ex}}$  has both requested properties.

## Provenance and scope

This is a literature-already-answered identification, not a new counterexample. The question appears in arXiv:1803.05052 (submitted March 2018), and the explicit negative answer appears in arXiv:1804.05730 (April 2018). The packet is restricted to Question 3. It makes no claim about the other four questions in Section 6 of the source.

## References

- [1] P. M. Berná, S. J. Dilworth, D. Kutzarova, T. Oikhberg, and B. Wallis, *The weighted Property (A) and the greedy algorithm*, arXiv:1803.05052.
- [2] P. M. Berná, *Equivalence between almost-greedy and semi-greedy bases*, arXiv:1804.05730.
- [3] P. M. Berná, Ò. Blasco, and G. Garrigós, *Lebesgue inequalities for the greedy algorithm in general bases*, Rev. Mat. Complut. 30 (2017), 369–392.